PROPOSAL

To

WORLD HEALTH ORGANIZATION AFRICA REGIONAL OFFICE'S NETWORK OF PARTNER INSTITUTIONS INITIATIVE:

PILLAR 2: ENHANCING HEALTH WORKFORCE DEVELOPMENT FOR SEXUAL AND REPRODUCTIVE HEALTH AND RIGHTS (SRHR)

BY

DEPARTMENT OF STATISTICS AND ACTUARIAL SCIENCES (STACS), JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY (JKUAT)

OCTOBER 2025

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1. Introductoin

This proposal is submitted with full institutional endorsement (Ref: Request for Institutional Letter of Support, see Appendix 1), affirming our commitment to contributing to Pillar 2 through: Development of regional education standards for health professionals and Operational research to inform policy and practice in workforce retention, deployment, and productivity.

Through this collaboration under Pillar 2, we aim to develop scalable, Africa-centred solutions for health workforce development with a sustained focus on SRHR. We bring institutional capacity, technical expertise, and strategic networks to accelerate regional progress in capacity-building, knowledge translation, and health systems leadership. We view this opportunity not only as a contribution to continental transformation - but as an imperative for shared resilience and human dignity. We are supported by institutions of interest

1.2 Institutional Profiles

1.2.1 Department of Statistics and Actuarial Sciences (STACS), Jomo Kenyatta University of Agriculture and Technology (JKUAT)

Established in 1981 Jomo Kenyatta University of Agriculture and Technology (JKUAT) is governed by the Universities Act – the laws of Kenya (2012). JKUAT is a top-tier public university committed to advancing knowledge through teaching, research, innovation, entrepreneurship and community engagement. With a strong emphasis on science, technology, engineering, and mathematics, the university serves over 35,000 students across 6 colleges and 23 schools.

The Department of Statistics and Actuarial Sciences (STACS), housed in the College of Pure and Applied Sciences (COPAS), spearheads transformative research and capacity-building initiatives in health systems - particularly in SRHR workforce analytics, deployment modelling, and ROI analysis. Our involvement in the World Health Organization Regional Office for Africa (WHO AFRO) Partner Institutions Network reflects JKUAT's strategic mandate to lead regionally in health systems strengthening.

1.3 Institutional Capacity

1.3.1 JKUAT's positioning

Institutional Capacity and Readiness: JKUAT satisfies all eligibility criteria for participation in Pillar 2, offering:

1. Technical Expertise - National leader in market-driven courses in human resources development, Statistics, Operational Research, and financial engineering, that will deliver on health workforce modelling and statistical forecasting.

2. Strategic Partnerships - Established collaborations and memoranda of understanding with public sector, private sector and regulatory institutions.
3. Infrastructure - Dedicated data labs, e-learning systems, and cross-disciplinary faculty teams
4. Gender Equity Focus - SRHR components integrated across nursing, pharmacy, and public health programs. The institution has a gender mainstreaming policy and a gender office.

JKUAT stands ready to contribute to the WHO AFRO Network of Partner Institutions by offering cutting-edge analytics, scalable training infrastructure, and strategic policy engagement. Our participation will generate high-impact data and actionable insights to shape national and regional HRH agendas across Africa.

1.3.2 Our team

i. Prof Anthony Waititu, Dean School of Mathematical and Physical Sciences (SMPS)
ii. Dr. Charity Wamwea, Chair of the Statistics and Actuarial Sciences Department (STACS)
iii. Dr. Joseph K. Mung'atu (Statistical modelling, STACS)
iv. Dr. Jane A. Akinyi (Statistician and health researcher, STACS)
v. Dr. Winnie Chacha (financial modeller and statistician, STACS)
vi. Dr. Ann Omamo (Human Resource Management, JKUAT)
vii. Mr. Henry Kissinger Ochieng (Statistician, STACS)
viii. Ms. Caro W. Mugo (Statistician, STACS)
ix. Dr. Matabel Odin (Financial Mathematics, STACS)

2. Background And Rationale

Despite important strides in expanding access to sexual and reproductive health and rights (SRHR) across Africa, the continent continues to face profound and persistent challenges in the development, deployment, and retention of a fit-for-purpose health workforce. These challenges include severe workforce shortages, inconsistent and fragmented training standards, underinvestment in leadership and governance capacities, and the insufficient integration of SRHR competencies into education and health systems. Collectively, these gaps jeopardise progress toward Universal Health Coverage (UHC), the Sustainable Development Goals (SDGs), and the realisation of equitable, rights-based SRHR outcomes (WHO, 2022a; AfHRH, 2023; WHO, 2023).

The increasing production capacity (255000 health workers yearly) of the health professions education institutions in the African region in addition to increasing recruitment have improved key health workforce indicators over that last decade (Asamani et al., 2024). Ironically, the leap in the production of health workers in the African region only reduced the shortage of health workers by 7% compared to the 33% reduction in the shortage of health workers globally (Boniol et al., 2022). Also, more than 23% of students enrolled in the health professions education/training fail to graduate, creating deficiencies in the health worker production and availability (Asamani et al., 2024). Further, about 27% of the health workers produced are employed, creating a paradox of shortage and unemployment attributed to poor HRH management and fiscal space deficits in member countries.

Beyond the critical shortage and unemployment of health workers is the issue of distribution and capacity (competency) to provide the appropriate healthcare services to the population. For example, only about 30% of the countries in Africa indicated using competency-based curriculum in the training of health professionals. Coincidentally, research has shown that about 40% of the health professionals in Africa are likely to miss diagnose patients they consult and out of the 64% who provide accurate diagnosis, less than 50% provide appropriate treatment (Di Giorgio et al., 2020). It is much concerning that sexually transmitted diseases and reproductive health associated significantly contribute to mortality and morbidity. Africa and other low-income regions are battling with a disproportionately high maternal mortality (346 deaths per 100000 live births) compared to 10 per 100000 in the other regions (World Health Organization, 2025). Likewise, in 2023, Africa contributed 70% of maternal deaths globally, leaving women and children of reproductive age at a very high risk of death (1 in 66) compared with 1 in 7933 in high income countries.

In the SRHR domain, these health workforce challenges are particularly pronounced. Gaps in pre-service and in-service training, both in curriculum content and delivery, contribute to inadequate competencies in core areas such as adolescent-responsive services, dignified maternity care, family planning, post-abortion care, and prevention of gender-based violence (UNFPA, 2022; WHO, 2023).

The COVID-19 pandemic exposed further vulnerabilities in workforce preparedness, resilience, and surge capacity. It also highlighted long-standing structural weaknesses in employment conditions, professional regulation, and information systems. Fewer than half of African countries have functioning National Health Workforce Accounts (NHWAs) or health workforce observatories, and less than one-third maintain updated HRH registries (WHO AFRO, 2022). Inefficiencies in resource use are also persistent, with many countries facing the paradox of graduate unemployment amidst workforce shortages due to fiscal constraints or lack of workforce absorption plans (UN HLC, 2016).

The crisis is not new. The *World Health Report 2006* identified the African Region as having the most acute global shortage of health workers, with 36 of the 57 globally affected countries located in the region (WHO, 2006). This triggered the World Health Organization to institute various health workforce reforms and lead many initiatives targeted at producing the adequate numbers of competent, optimal mix, well-motivated, skilfully distributed and supported health workforce to meet the demands of universal health coverage (Asamani et al., 2024). The adoption of the *Regional Roadmap for Scaling Up Human Resources for Health for Improved Health Service Delivery (2012–2025)*, and subsequent global and regional initiatives, development of the health workforce investment charter, establishment of the Curriculum Development Advisory Committee, the Health Professions Education Dialogue and commissioning of the competency-based prototype curricula are notable health professions education interventions, some of which have started making imparts

However, despite some progress, most countries in the Region remain well below the WHO-recommended minimum threshold of 2.3 doctors, nurses, and midwives per 1,000 population

(WHO, 2023), with vast rural–urban disparities and an unmet need for skilled health workers, particularly in SRHR services. In recognition of these persistent gaps, the *Global Strategy on Human Resources for Health: Workforce 2030 (GSHRH),* adopted through World Health Assembly (WHA) resolution 69.19, has become the blueprint for Human Resources for Health (HRH) reform. It calls on Member States to:

1. Optimize the performance, quality, and impact of the health workforce.
2. Align investment with population and health system needs.
3. Strengthen governance and institutional capacity for HRH stewardship; and
4. Improve data systems for accountability and planning (WHO, 2016).

These objectives have been contextualised for the African Region through the WHO AFRO *Regional Strategy on Human Resources for Health 2022–2030*, which prioritises institutionalising education quality standards, scaling leadership and management capacity, and investing in operational research and data systems for health workforce productivity (WHO AFRO, 2022).

Yet, there is strong evidence that investing in HRH yields a triple return: improved health outcomes, enhanced global health security, and economic growth (UN HLC, 2016). Every $1 invested in the general health workforce can return up to $9 in economic gains, an even higher return when focused on community health workers (WHO, 2022). The time to scale such investment is now.

To reverse the ineffectiveness and inefficiencies in the African health systems coupled with the uncertainties in donor funding, HRH crisis in Africa and to ensure SRHR readiness, a transformative, multisectoral, evidence-informed response is needed; one that prioritises:

i. The development and institutionalization of context-relevant, quality-assured standards for health professions education/training (pre-service and in-service) will add to the existing initiatives to strengthen and enhance the development of health workforce to ensure consistency, excellence, and relevance, including health workforce for Sexual and Reproductive Health Rights.
ii. Operational research on workforce needs, deployment, and productivity, including tools for continuing professional development (CPD), promoting gender-equitable, inclusive, and rights-based workforce policies.

3. Technical Approach

JKUAT will lead the design and implementation of a comprehensive research and knowledge translation agenda focused on identifying and addressing workforce gaps related to health worker availability, distribution, retention, deployment, and CPD. This will generate actionable evidence and decision-support tools to inform policies and investments for optimizing the health workforce for SRHR service delivery across the African Region.

4. Methodology

The team will employ a multi-method, data-driven approach - combining quantitative modelling (e.g., regression analysis, scenario simulations), qualitative stakeholder engagement (interviews, workshops), and digital tool development (dashboards, algorithms) - to assess, forecast, and optimize SRHR workforce capacity in the African Region, while ensuring policy relevance through iterative stakeholder collaboration and knowledge translation. Additionally, the consortium will support research and knowledge translation on workforce needs, planning, and productivity, including developing tools for workforce deployment, retention, and continuing professional development. Table 1 provides a summary of the research methods.

Table 1: Summary project plan

Phase	Objective	Methods/Mechanisms	Sample	Data Collection & Analysis	Stakeholder Involvement & Evaluation	Expected Output	Cost Implications (USD)
Evidence Synthesis	Assess workforce status and gaps in SRHR across the African Region to inform evidence-based planning	Mixed-methods: Desktop reviews, stakeholder interviews, data triangulation; Workforce stock/density mapping; Training pathway analysis	HRH policymakers, SRHR providers, training institutions; NHWA, HRH registries, policy documents	Attrition/deployment trends; Equity-focused demographic analysis	Ministries, professional councils, and frontline workers engaged to validate gaps and priorities	Baseline for regional workforce planning; identification of critical shortages and systemic barriers	5,000.00
Development and validation	Quantify workforce productivity and training impact on SRHR outcomes	ROI modelling (economic and health outcomes); Regression analysis of training efficacy; DALY-linked workforce investment models	Facility reports, training records, national health indicators	Productivity ratios (services/provider); Training-to-performance linkages; Human capital gains estimation	Ministries and training institutions validate findings for policy integration	ROI evidence to inform HRH investment priorities (e.g., CPD expansion)	9,200.00
Dissemination	Translate evidence into policy action and regional knowledge-sharing	Policy briefs; Peer-reviewed publications; Infographics; Stakeholder workshops; Open-access repositories	Ministries, WHO, NGOs, academia	Synthesized reports with actionable recommendations; Dissemination metrics (reach, uptake)	Multi-stakeholder workshops ensure buy-in and adaptation to local contexts	Final policy products; Cross-country collaboration on SRHR workforce solutions	3,237.50
Administration cost (Grant management) @ 7%							1,312.50
Total							**18,750.00**

Table 2: Workplan with the milestones

Phase	Key Activities	Timeline	Milestones/Deliverables
Evidence Synthesis	Review SRHR workforce data and literature	Nov 1st - Nov 30, 2025	SRHR workforce gap analysis
	Map workforce stock and training pathways		WHO feedback received
Development	Analyze attrition, deployment, and equity trends	Dec 1 – 15 December, 2026	
	Document and analyse consensus inputs	Dec 1 - Jan 15, 2028	WHO review of stakeholder list and draft
Analysis	Model ROI of HRH investments	Jan 5 - Feb 25, 2026 (overlaps)	ROI report finalized
	Conduct regression analysis of training efficacy		Policy-relevant productivity metrics validated by ministries
	Estimate DALY-linked human capital gains		
Dissemination	Draft policy briefs and peer-reviewed publications	Feb 15 - Mar 15, 2026	Final policy products published
	Develop infographics and open-access resources		Africa Health Professions Education Quality Standards disseminated
	Host stakeholder workshops for cross-country knowledge sharing		
Closure	Submit final project report to WHO for review and publication	End of March 2026	WHO-reviewed report submitted
Administration	Grant management, reporting, and coordination	Ongoing (Nov - Mar 2026)	Timely financial and narrative reports submitted
			Disbursements tracked and reconciled

5. Evaluation

The project will be evaluated iteratively, and in line with the deliverables in Table 2.

6. Budget

Table 3: The budget summary

Phase	**Cost Implications (USD)**
Evidence Synthesis	5,000.00
Development and validation	9,200.00
Dissemination	3,237.50
Administration cost (Grant management) @ 7%	1,312.50
Total	**18,750.00**

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